



M-Tech Production & Marketing

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Experiment 3

LAB TASKS

Internship Program — Batch 2026

Submitted By:	Abdullah Rafique (MTech-AI260120)
Department:	AI / ML
Submitted To:	Ceo Moazzan
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1. Experiment 3 – Lab Tasks

File: *Experiment_3.py*

2. Viva Questions

1. Why is a NumPy array faster and more memory-efficient than a Python list?

A NumPy array is significantly faster and more memory-efficient than a standard Python list because it stores elements of the same data type in contiguous memory locations. This allows NumPy to perform computations more efficiently. Additionally, NumPy supports **vectorized operations**, enabling mathematical calculations on an entire array without using explicit loops. It also provides optimized built-in functions such as `sum()`, `mean()`, `std()`, `min()`, and `max()`, making numerical computing faster and more efficient.

2. What does the shape of a NumPy array represent?

The **shape** of a NumPy array describes its dimensions by indicating the number of rows and columns. For example, an array with a shape of **(3, 4)** contains **3 rows** and **4 columns**. Understanding an array's shape is essential when performing operations such as reshaping, indexing, slicing, and matrix multiplication.

3. What is the purpose of `axis=0` in NumPy?

In a two-dimensional NumPy array, **`axis=0`** performs operations **column-wise**. For example, when using the `sum()` function with `axis=0`, NumPy adds the values in each column and returns the sum of every column.

For a 3×4 array:

- **`axis = 0`** → Calculates the sum of each **column**.
- **`axis = 1`** → Calculates the sum of each **row**.

This concept is useful for performing statistical and mathematical operations on multidimensional data.

4. Why are vectorized operations faster than using loops?

Vectorized operations are faster because they are implemented in optimized **C language** within the NumPy library. Instead of processing elements one at a time using Python loops, NumPy performs calculations on the entire array simultaneously. This significantly reduces execution time and improves performance, especially when working with large datasets.

5. What is a Pandas DataFrame, and how is it different from a NumPy array?

A **Pandas DataFrame** is a two-dimensional, table-like data structure with labeled rows and columns, similar to a Microsoft Excel spreadsheet. Unlike NumPy arrays, which generally store a single data type, a DataFrame can store multiple data types such as integers, floating-point numbers, strings, and dates within different columns.

DataFrames also provide advanced features including:

- Data filtering
- Sorting
- Grouping
- Merging datasets
- Handling missing values
- Reading and writing CSV or Excel files

These capabilities make Pandas the preferred library for data analysis and preprocessing.

6. How can missing values be handled in Pandas?

Missing values are common in real-world datasets and can be managed in two primary ways:

- **Removing missing values** using the `dropna()` method.
- **Replacing missing values** using the `fillna()` method with appropriate values such as the mean, median, mode, or a custom value.

Examples:

```
df.dropna()
```

Removes rows containing missing values.

```
df.fillna(df.mean())
```

Replaces missing values with the mean of each column.

Proper handling of missing values improves data quality and ensures accurate analysis and machine learning results.

3. Multiple Choice Questions (MCQs)

Q1. The shape of a NumPy array with 3 rows and 4 columns is:

Answer: (B) (3, 4)

Q2. Which Python library is primarily used for working with tabular data?

Answer: (B) Pandas

Q3. What does `df.head()` display?

Answer: (B) The first rows of the DataFrame

Q4. A missing value in Pandas is represented as:

Answer: (B) NaN

Q5. In a two-dimensional NumPy array, `axis = 0` performs operations on:

Answer: (B) Columns

4. Intern Information

Field	Details
Intern Name	Abdullah Rafique
Supervisor Name	Mr. Mohazin Zahid
Date	06 July, 2026
Domain	AI/ML Intern
Intern Signature	<u>ABD</u>